

# Team Work Division — Global Farmer Platform P0

|                  |                                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------------|
| Source spec      | 6182.docx — Global Farmer Platform, Complete Product and Development Specification v1.0.0 (16 Sep 2026) |
| Baseline audited | main @ 000d2da , 17 Sep 2026                                                                            |
| Team             | 6 engineers, full time                                                                                  |
| Cadence          | 2-week sprints, 12 sprints (~24 weeks) to a 20-farm pilot                                               |
| Scope            | <b>P0 only:</b> PR-01 – PR-12, PR-25 – PR-27                                                            |
| Cloud            | <b>Last.</b> Everything runs on the local stack until Phase 5                                           |

Spec references in this document use the spec's own IDs: chapters ( §04 ), requirements ( PR-05 ), workflows ( W09 ) and acceptance scenarios ( T10 ).

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# 1. Where we are: codebase audit

All 11 tested Go packages pass. The code is a well-built prototype of **one analysis screen plus onboarding**. The spec describes a multi-tenant platform. Roughly **10–15% of P0 is done**, and the done part (Earth Engine from Go) is the spec's biggest technical risk.

## 1.1 Keep and build on

| Area                            | Location                                                                                                          | Size                  | Why it carries over                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------|
| Earth Engine expression builder | <code>internal/gee/eeexpr</code>                                                                                  | 1.1k lines + fixtures | Solves "Go EE REST expression proof" (§02.7 gate, §04.1)   |
| Analysis pipeline               | <code>internal/pipeline</code>                                                                                    | 1.7k lines            | Sentinel-2/Sentinel-1 collections, index math, grid, stats |
| Earth Engine client and session | <code>internal/gee</code>                                                                                         | 670 lines             | Auth, paging, retry on 429/5xx only, token refresh fixed   |
| Polygon validation              | <code>internal/geo</code>                                                                                         | 200 lines, tested     | Basis for T08                                              |
| Middleware                      | <code>internal/httpapi/middleware</code>                                                                          | tested                | Rate limit, body cap, in-flight cap, request IDs           |
| Config pattern                  | <code>internal/config</code>                                                                                      | tested                | Single tuning surface stays                                |
| Chatbot                         | <code>internal/chatbot</code> , <code>internal/gemini</code>                                                      | prototype             | Basis for PR-17 after the pilot                            |
| Frontend pieces                 | <code>HeatmapLayer.jsx</code> , <code>Legend.jsx</code> , <code>colorUtils.js</code> , <code>pages/steps/*</code> | —                     | UX reference to port into Next.js                          |

## 1.2 Gaps against the spec

| Area                   | Today                                                                                                                                   | Spec requires                                                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Process model          | One Gin server; EE analysis runs inside the HTTP request (300 s write timeout)                                                          | <code>cmd/api</code> , <code>cmd/worker</code> , <code>cmd/tile</code> ; SQL jobs; outbox; no work after the response (§02.4–5) |
| Persistence of results | <b>None.</b> <code>vi_reports</code> is never written (AUDIT L3)                                                                        | Immutable observations + latest projection (§04.3, §07.4)                                                                       |
| Database               | 8 farmer-centric tables; <code>schema.sql</code> and <code>mindstrix_setup.sql</code> disagree; no migration runner (AUDIT M4); M2 open | ~43 tables, tenant-scoped, RLS FORCE (§07, §17, §18)                                                                            |
| Identity               | Phone OTP + custom JWT + Werkzeug hashes                                                                                                | Identity Platform tokens; memberships, roles, field grants (§09)                                                                |
| API                    | 24 unversioned routes; 3 error envelopes; <code>/api/sample</code> global slot (AUDIT H7)                                               | 104 <code>/v1</code> endpoints; problem+json; ETag; Idempotency-Key; signed cursors (§06.2–3, §15)                              |

| Area        | Today                                                                                                                                                           | Spec requires                                                                                                            |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Science     | SCL mask; NDVI/EVI/SAVI/NDMI/NDWI/GNDVI/CVI; mean + std only; Gaussian-smoothed grid                                                                            | Cloud Score+; NDVI/NDMI/NDRE/RECI/MSAVI/EVI at native resolution; valid fraction, p10/p50/p90; COG + tile gateway (\$04) |
| Tests       | 0 tests in <code>repo</code> , <code>db</code> , <code>firebase</code> , <code>firestore</code> , <code>gemini</code> , <code>jwtutil</code> , <code>cmd</code> | SQL integration, isolation, E2E, performance (\$12.1)                                                                    |
| Frontend    | Vite + React + Leaflet, 5.9k lines                                                                                                                              | Next.js <code>output: 'export'</code> , 17 routes, offline scouting (\$08)                                               |
| Not started | —                                                                                                                                                               | Import, seasons/crop cycles, weather, alerts, scouting, activities, reports, notifications, entitlements, privacy, audit |
| DevOps      | Dockerfile only; no CI; JSON key file                                                                                                                           | CI, migrations, Terraform, WIF (Phase 5)                                                                                 |

## 2. Ground rules

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1. **The frozen-contract rule in `CLAUDE.md` is lifted.** The new `/v1` API and `apps/web` are built alongside the legacy app. The legacy app keeps running as a reference and is deleted at the end of Phase 4.
  2. **PO only.** Anything in §13 of this document waits for the pilot.
  3. **Cloud last.** No one provisions or depends on GCP before Phase 5. Every cloud service sits behind a Go interface with a local implementation (§3).
  4. **Contract first.** `contracts/openapi.json` is frozen at the end of Sprint 2. Frontend builds against the mock server until endpoints land.
  5. **Earth Engine and Gemini are called live** from dev machines (they are external APIs, not our infrastructure). Every new EE call gets a recorded fixture.
  6. **Owners decide inside their folders.** Cross-folder changes need the other owner's review (CODEOWNERS).
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### 3. Local stack (replaces the cloud until Phase 5)

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Owned by **P1**. Ready by the end of Sprint 1. `docker compose up` must start everything.

| Spec component (Phase 5)                   | Local implementation                            | Go interface                     |
|--------------------------------------------|-------------------------------------------------|----------------------------------|
| Cloud SQL PostgreSQL + PostGIS             | <code>postgis/postgis</code> container          | <code>pgxpool</code> (unchanged) |
| Memorystore Redis                          | <code>redis</code> container                    | <code>cache.Store</code>         |
| Cloud Storage (quarantine / ready buckets) | MinIO or <code>fake-gcs-server</code>           | <code>storage.Bucket</code>      |
| Pub/Sub + Cloud Tasks                      | <code>jobs</code> table + in-process dispatcher | <code>queue.Dispatcher</code>    |
| Identity Platform                          | Firebase Auth emulator                          | <code>auth.Verifier</code>       |
| Secret Manager                             | <code>.env</code>                               | <code>config</code>              |
| Google Weather API                         | Open-Meteo adapter                              | <code>weather.Provider</code>    |
| Cloud Logging / Trace                      | stdout slog + local OTel collector (optional)   | <code>telemetry</code>           |

Rule: **no domain package imports a concrete adapter**. Phase 5 swaps implementations without touching domain code.

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## 4. Team and ownership

| ID | Role                                              | Primary requirements                                               | Owns (CODEOWNERS)                                                                                                                                                                                                                                                                                |
|----|---------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P1 | Tech lead / platform backend                      | PR-01, PR-27 (platform parts)                                      | <code>cmd/api</code> , <code>internal/platform/**</code> , <code>internal/domain/{tenancy,identity,audit}</code> , <code>internal/adapters/{postgres,redis,storage,queue}</code> , <code>migrations/</code> , <code>contracts/</code> , <code>docker-compose.yml</code> , <code>AGENTS.md</code> |
| P2 | Geospatial engineer                               | PR-05, PR-06 (data), PR-08 (detection)                             | <code>internal/gee/**</code> , <code>internal/pipeline/**</code> , <code>internal/domain/observations</code> , <code>cmd/worker/analysis</code> , <code>cmd/tile</code>                                                                                                                          |
| P3 | Backend — fields and farm data                    | PR-02, PR-03, PR-04, onboarding, legacy data                       | <code>internal/domain/{farms,fields,seasons,imports,soil,consent}</code> , <code>tools/migrate-legacy</code>                                                                                                                                                                                     |
| P4 | Backend — operational workflows                   | PR-07, PR-08 (lifecycle), PR-09, PR-10, PR-11, PR-12, PR-25, PR-26 | <code>internal/domain/{weather,alerts,scouting,activities,reports,notifications,usage,privacy}</code> , <code>internal/adapters/weather</code>                                                                                                                                                   |
| P5 | Frontend lead                                     | PR-06 UI, UI for PR-01 – PR-08                                     | <code>apps/web/**</code> except the folders owned by P6                                                                                                                                                                                                                                          |
| P6 | Frontend + QA/DevEx, <b>cloud lead in Phase 5</b> | UI for PR-09 – PR-12, PR-26; PR-27 (release, cloud)                | <code>apps/web/src/features/{scouting,activities,reports,settings,privacy}</code> , <code>apps/web/src/offline</code> , <code>tests/**</code> , <code>.github/**</code> , <code>infra/**</code>                                                                                                  |

**Tie-breaks:** P1 decides architecture and contract questions. P2 decides scientific method questions. The whole team decides scope changes in sprint review.

## 5. Phase plan

| Phase               | Sprints | Weeks | Goal                                                                          | Exit gate (all must pass)                                                                                                |
|---------------------|---------|-------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 0 — Foundation      | S1      | 1–2   | New layout, local stack, contract draft, science proof                        | <code>docker compose up</code> green; CI running; Cloud Score+ fixture passes; old/new table map approved                |
| 1 — Platform core   | S2–S3   | 3–6   | Tenancy, identity, RLS, fields, jobs framework                                | T01, T02, T03, T08 pass; OpenAPI frozen (end S2); field created via <code>/v1</code> from the new UI                     |
| 2 — Monitoring      | S4–S6   | 7–12  | Observations, maps, seasons, import, weather, alerts                          | T05, T09, T10–T14 pass; 1,000-field local benchmark; field detail shows persisted data                                   |
| 3 — Workflows       | S7–S9   | 13–18 | Scouting (offline), activities, reports, notifications, entitlements, privacy | T15–T18, T20, T25, T30 pass; W01–W16, W21, W23, W24, W28 demoed end to end                                               |
| 4 — Hardening       | S10     | 19–20 | Security, accessibility, performance, legacy migration, legacy removal        | All P0 T-scenarios green locally; legacy data migrated twice on a copy; <code>frontend/</code> and legacy routes deleted |
| 5 — Cloud and pilot | S11–S12 | 21–24 | Terraform, CI/CD, staging, production, DR, pilot                              | Staging and production live; restore drill passed; T26–T29 pass in staging; 20 pilot farms onboarded                     |

**Critical path:** OpenAPI + tenancy/RLS (P1, S2) → fields (P3, S2–S3) → observation jobs (P2, S4–S5) → map & alerts (P5/P4, S5–S6) → scouting & activities (P4/P6, S7–S8) → cloud (P6/P1, S11–S12).

## 6. Sprint board

| Sprint                | P1 Platform                                                               | P2 Geospatial                                                | P3 Fields                                 | P4 Workflows                                        | P5 Frontend                                     | P6 Frontend + QA                                                    |
|-----------------------|---------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------|-----------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------|
| <b>S1</b> (wk 1–2)    | Repo layout, local stack, AGENTS.md, OpenAPI draft                        | Cloud Score+ proof, fixtures, index spec                     | Legacy → new table map                    | Weather adapter (Open-Meteo), alert rule spec       | Next.js static shell, routes, tokens            | CI, test harness, isolation suite skeleton                          |
| <b>S2</b> (wk 3–4)    | Tenancy migrations, RLS, auth verifier, tx pattern. <b>OpenAPI frozen</b> | NDRE/RECI/MSAVI, 20 m handling, per-field stats              | Farms, fields, boundary versioning        | Weather storage and field bindings                  | API client from OpenAPI, sign-in, tenant switch | T01/T02 tests, fill untested legacy packages                        |
| <b>S3</b> (wk 5–6)    | Idempotency, ETag, cursors, audit, outbox, jobs + leases                  | Quality classes (usable/partial/insufficient)                | Field groups, archive, field API complete | Entitlements & usage tables                         | Field list, MapLibre, draw/edit                 | T03/T08 tests, E2E: sign-in → field                                 |
| <b>S4</b> (wk 7–8)    | Queue dispatcher, storage adapter, reaper                                 | Pipeline in worker jobs, EE adapter ops                      | Seasons, crop cycles, overlap lock        | Admission checks before costly work                 | Scene timeline, index chart                     | E2E harness vs exported build                                       |
| <b>S5</b> (wk 9–10)   | Membership & invitation API                                               | COG export, manifests, observations, latest projection       | Import: upload, dry-run                   | Alert engine: episodes, dedupe, data-gap            | Field detail, map tiles, quality/source panel   | T09/T10/T12 tests, benchmark harness                                |
| <b>S6</b> (wk 11–12)  | Contract review, security review of S1–S5                                 | Tile gateway, change detection, <b>1,000-field benchmark</b> | Import commit, resume, dedupe             | Alert triage, acknowledgement                       | Home work list, alerts, weather, compare dates  | T11/T13/T14 tests, import wizard UI                                 |
| <b>S7</b> (wk 13–14)  | Audit queries, admin endpoints                                            | Processing versions, reprocessing rules                      | Soil tests, consent records               | Scout tasks, inspections                            | Team & invitations UI                           | Offline storage (IndexedDB), sync queue                             |
| <b>S8</b> (wk 15–16)  | Privacy job framework                                                     | Backfill limits, numeric goldens                             | New onboarding flow API                   | Offline sync API, activities & costs                | Onboarding UI on new API                        | Scouting UI, conflict screen, activities UI                         |
| <b>S9</b> (wk 17–18)  | Retention jobs, cursor hardening                                          | Tune profiles, cost accounting                               | Legacy migration script v1                | Reports worker, notifications, privacy export/erase | Accessibility pass, English/Hindi               | Reports & settings UI, T15–T18/T20 tests                            |
| <b>S10</b> (wk 19–20) | Security fixes, final contract, remove legacy routes                      | Science sign-off, docs                                       | Legacy migration rehearsal ×2             | Bug fixes, T25/T30                                  | Performance, delete frontend/                   | Full regression, local load test, release checklist                 |
| <b>S11</b> (wk 21–22) | Swap adapters to Cloud SQL/Tasks/GCS/Identity Platform                    | EE service account & quota verification in staging           | Migrate legacy data into staging          | Google Weather adapter in staging                   | Firebase Hosting config, CSP headers            | <b>Terraform staging</b> , WIF, CI/CD deploy, monitoring            |
| <b>S12</b> (wk 23–24) | Production readiness review                                               | Production data-quality check                                | Pilot farm onboarding support             | Notification provider in production                 | Pilot UI fixes                                  | <b>Terraform production</b> , restore drill, T26–T29, pilot rollout |

## 7. Detailed work by person

Each task lists **what**, **where**, **done when** and **spec**.

### P1 — Tech lead / platform backend

**Mission:** everyone else builds on your contract, transaction pattern and adapters. In S1–S2 you do nothing else.

#### Phase 0 (S1)

| #   | Task                                                                                                       | Where                                                                                    | Done when                                                                        | Spec            |
|-----|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------|
| 1.1 | New repository layout; keep legacy under <code>cmd/server</code> + <code>internal/httpapi</code> until S10 | <code>cmd/{api,worker,tile}</code> ,<br><code>internal/{platform,domain,adapters}</code> | <code>go build ./...</code> passes; legacy <code>make verify</code> still passes | §06.1           |
| 1.2 | Choose and pin <code>chi</code> , <code>goose</code> , <code>sqlc</code> ; record as ADRs                  | <code>docs/adr/</code>                                                                   | ADR merged                                                                       | §06.1, §13      |
| 1.3 | Local stack                                                                                                | <code>docker-compose.yml</code>                                                          | One command starts PostGIS, Redis, MinIO, Firebase emulator, API, worker         | §3 of this doc  |
| 1.4 | <code>AGENTS.md</code> rules for Antigravity                                                               | repo root                                                                                | Merged; every engineer has read it                                               | §11 of this doc |
| 1.5 | OpenAPI draft for PR-01 – PR-12, PR-25 – PR-27 resources from spec §14 and §15                             | <code>contracts/openapi.json</code>                                                      | Validates; mock server (Prism) runs                                              | §06.2, §14, §15 |
| 1.6 | Decide the fate of <code>schema.sql</code> vs <code>mindstrix_setup.sql</code>                             | <code>docs/adr/</code>                                                                   | ADR merged                                                                       | AUDIT M4        |

#### Phase 1 (S2–S3)

| #   | Task                                                                                                                                                                                                                                                                                                                             | Where                               | Done when                             | Spec       |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|---------------------------------------|------------|
| 1.7 | Migrations: <code>tenants</code> , <code>memberships</code> , <code>invitations</code> , <code>membership_field_grants</code> , <code>authorization_revisions</code> , <code>audit_records</code> , <code>outbox</code> , <code>inbox</code> , <code>idempotency_records</code> , <code>jobs</code> , <code>job_execution</code> | <code>migrations/</code>            | Up/down from empty DB in CI           | §17, §18   |
| 1.8 | RLS: <code>ENABLE</code> + <code>FORCE</code> on tenant tables; runtime role without <code>BYPASSRLS</code>                                                                                                                                                                                                                      | <code>migrations/</code>            | T01 isolation suite passes            | §07.3      |
| 1.9 | Token verifier (Firebase emulator locally) and principal resolution                                                                                                                                                                                                                                                              | <code>internal/platform/auth</code> | Invalid → 401; suspended member → 403 | §09.3, W01 |

| #    | Task                                                                            | Where                                      | Done when                                                                           | Spec   |
|------|---------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------|--------|
| 1.10 | Transaction helper: <code>set_config('app.tenant_id', ..., true)</code> + actor | <code>internal/platform/db</code>          | No handler can query without it; test proves pooled connection does not leak tenant | \$06.6 |
| 1.11 | problem+json errors, request ID, body cap (reuse middleware)                    | <code>internal/platform/http</code>        | All <code>/v1</code> errors match schema                                            | \$06.2 |
| 1.12 | <b>Freeze OpenAPI</b> (end S2)                                                  | <code>contracts/</code>                    | Tagged <code>contract-v1.0.0-rc1</code>                                             | \$06.2 |
| 1.13 | Idempotency-Key reservation in-transaction                                      | <code>internal/platform/idempotency</code> | T03, T04 pass                                                                       | \$06.3 |
| 1.14 | ETag / If-Match, version increment                                              | <code>internal/platform/http</code>        | Stale If-Match → 412                                                                | \$06.2 |
| 1.15 | HMAC-signed keyset cursors                                                      | <code>internal/platform/pagination</code>  | Revocation invalidates cursor                                                       | \$06.2 |
| 1.16 | Audit + outbox rows written in the same transaction                             | <code>internal/domain/audit</code>         | Every write endpoint emits both                                                     | \$02.4 |
| 1.17 | Jobs with SQL leases and fence tokens                                           | <code>internal/platform/jobs</code>        | Only current lease holder can publish                                               | \$02.5 |

### Phase 2–3 (S4–S9)

| #    | Task                                                         | Done when                                                     | Spec   |
|------|--------------------------------------------------------------|---------------------------------------------------------------|--------|
| 1.18 | <code>queue.Dispatcher</code> local implementation + reaper  | Killed worker's job is re-dispatched once (T07)               | \$02.5 |
| 1.19 | <code>storage.Bucket</code> with quarantine/ready separation | Upload cannot reach ready without validation (T17 groundwork) | \$07.6 |
| 1.20 | Memberships, invitations, field-grant API                    | W02, W03 demoed; T02 passes                                   | PR-01  |
| 1.21 | Security review of every domain package merged in S1–S5      | Findings filed and closed                                     | \$09   |
| 1.22 | Privacy job framework (used by P4)                           | Erasure job plan persisted and resumable                      | \$07.7 |
| 1.23 | Retention jobs (idempotency 24 h, inbox 30 d, logs policy)   | Jobs run and are tested                                       | \$07.7 |

### Phase 4–5 (S10–S12)

| #    | Task                                                                               | Done when                        |
|------|------------------------------------------------------------------------------------|----------------------------------|
| 1.24 | Remove legacy routes, <code>internal/httpapi</code> , <code>cmd/server</code>      | Build green without them         |
| 1.25 | Swap adapters to Cloud SQL, Cloud Tasks, Pub/Sub, GCS, Identity Platform (with P6) | Staging runs the same test suite |
| 1.26 | Production readiness review                                                        | Release checklist signed         |

## P2 — Geospatial engineer

**Mission:** turn the synchronous heatmap prototype into a job-based, provenance-tracked observation pipeline that never reports missing data as healthy.

### Phase 0 (S1)

| #   | Task                                                                                                                   | Where                                                                      | Done when                                                             | Spec          |
|-----|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------|
| 2.1 | Cloud Score+ linking proof ( <code>GOOGLE/CLOUD_SCORE_PLUS/V1/S2_HARMONIZED</code> , <code>cs_cdf</code> $\geq 0.60$ ) | <code>internal/pipeline/sentinel2.go</code> , <code>eeexpr/testdata</code> | Fixture captured; live test under <code>GEE_LIVE_TEST=1</code> passes | \$04.2        |
| 2.2 | Rename <code>MaskCloudsSCL</code> path to an explicitly labelled degraded fallback                                     | <code>internal/pipeline</code>                                             | Output carries <code>mask_method</code>                               | \$04.3 step 5 |
| 2.3 | Write the index registry as config                                                                                     | <code>internal/config</code>                                               | Matches \$04.4 table exactly                                          | \$04.4        |

### Phase 1 (S2–S3)

| #   | Task                                                                                                      | Done when                                                 | Spec          |
|-----|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------|
| 2.4 | Add NDRE, RECI, MSAVI; keep NDVI, NDMI, EVI; drop CVI/SAVI/NDWI/GNDVI from <code>/v1</code>               | Formula tests with scaled reflectance (T12)               | \$04.4        |
| 2.5 | Aggregate N to 20 m for NDMI; red-edge indices on 20 m grid                                               | Effective resolution stored per result (T11)              | \$04.3 step 7 |
| 2.6 | Denominator epsilon 1e-6; invalid → masked, never Inf/0                                                   | T10 unit test: all-cloud → insufficient                   | \$04.4        |
| 2.7 | Per-field reducer: valid area fraction, pixel count, area-weighted mean, p10/p50/p90, population std      | Golden values for the 5 existing numeric fixture polygons | \$04.3 step 8 |
| 2.8 | Quality classes: usable ( $\geq 0.70$ , $\geq 20$ px), partial ( $\geq 0.30$ , $\geq 5$ px), insufficient | Classes returned on every observation                     | \$04.5        |

### Phase 2 (S4–S6)

| #    | Task                                                                                                                                                                 | Done when                                                      | Spec           |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|----------------|
| 2.9  | Move analysis out of HTTP into worker jobs ( <code>AnalysisRequest</code> → 202 + job)                                                                               | No EE call on the request path                                 | \$02.4, W09    |
| 2.10 | EE adapter: <code>DiscoverScenes</code> , <code>SubmitStatistics</code> , <code>SubmitRasterExport</code> , <code>GetOperation</code> , <code>CancelOperation</code> | Each has recorded fixtures incl. 429/5xx/timeout               | \$04.1         |
| 2.11 | Persist EE operation names; reconcile uncertain submissions                                                                                                          | T05 passes                                                     | \$02.5         |
| 2.12 | COG export to <code>storage.Bucket</code> with overviews, CRS, no-data mask, checksum, manifest                                                                      | Manifest validated before publish                              | \$04.3 step 9  |
| 2.13 | <code>observations</code> (immutable) + latest-observation projection (with P3 for FKs)                                                                              | Boundary edit mid-job does not update current projection (T09) | \$04.3 step 10 |
| 2.14 | Tile gateway: grant check per request, COG range reads, cache key tenant+digest+style+z/x/y                                                                          | Revoked user gets 404 on next tile                             | \$02.6         |

| #    | Task                                                                                                                             | Done when                                 | Spec   |
|------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------|
| 2.15 | Change detection on common valid pixels ( $\Delta\text{NDVI} \geq 0.15$ , gap 3–30 d, common valid $\geq 0.70$ ) → events for P4 | T13 passes                                | \$04.5 |
| 2.16 | Batching by tile/time window, concurrency 2, adaptive split                                                                      | <b>1,000-field local benchmark</b> report | \$04.7 |

### Phase 3–5 (S7–S12)

| #    | Task                                                                               | Done when                                                    |
|------|------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 2.17 | <code>processing_version</code> = graph digest + profile; reprocessing rules       | Two profiles coexist without collision                       |
| 2.18 | Historical backfill limits separate from fresh scenes                              | Backfill cannot starve fresh processing                      |
| 2.19 | Retire <code>smooth.go</code> from <code>/v1</code> (resampling is not new detail) | Maps show real rasters only                                  |
| 2.20 | Science documentation and sign-off                                                 | Index registry, quality policy, known limitations documented |
| 2.21 | S11: verify EE service account and batch quota in staging (with P6)                | Benchmark rerun in staging                                   |

## P3 — Backend: fields and farm data

**Mission:** permanent field identity, versioned boundaries, seasons and import — and move the existing farmers' data safely.

### Phase 0 (S1)

| #   | Task                                                                                                                                                                                                                                     | Done when                  | Spec               |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------------|
| 3.1 | Map legacy tables ( <code>farmers</code> , <code>farmer_locations</code> , <code>farms</code> , <code>crops</code> , <code>irrigation</code> , <code>soil_info</code> , <code>consents</code> , <code>vi_reports</code> ) to spec tables | Mapping doc approved by P1 | \$17               |
| 3.2 | Geometry fixture set: self-intersection, bad holes, swapped coords, antimeridian, >10,000 vertices, <0.01 ha, >20,000 ha                                                                                                                 | Fixtures committed         | T08, \$04.3 step 2 |

### Phase 1 (S2–S3)

| #   | Task                                                                                                                                         | Done when                      | Spec             |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------------------|
| 3.3 | Migrations: <code>farms</code> , <code>fields</code> , <code>field_boundaries</code> , <code>field_groups</code> , <code>group_fields</code> | With tenant FKs and GiST index | \$17, \$18       |
| 3.4 | Field create/edit: PostGIS validation, geodesic area (holes excluded), boundary row appended in the same transaction                         | Client-supplied area rejected  | \$07.2, W04, W06 |
| 3.5 | Reuse <code>internal/geo</code> checks before PostGIS conversion                                                                             | T08 passes with precise errors | T08              |
| 3.6 | Unique active field name within farm; archive flow                                                                                           | W07 demoed                     | PR-02            |
| 3.7 | Field groups CRUD                                                                                                                            | Grant filter respected         | PR-02            |

## Phase 2 (S4–S6)

| #    | Task                                                                                                                                          | Done when                                                     | Spec              |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------|
| 3.8  | <code>seasons</code> , <code>season_fields</code> , <code>crop_cycles</code> ; lock (tenant, field) for overlap checks                        | Overlapping same-field cycles rejected under concurrency test | §07.2, W08, PR-04 |
| 3.9  | Import upload via <code>storage.Bucket</code> grant; archive limits (100 MiB, 500 MiB extracted, 10,000 entries, traversal/symlink rejection) | Malicious archive fixtures rejected                           | §06.4             |
| 3.10 | Parsers: GeoJSON, KML, KMZ, SHP (zipped); reject ambiguous CRS                                                                                | Per-format fixtures                                           | W05               |
| 3.11 | Dry-run with per-row results and column mapping                                                                                               | UI can show every row outcome                                 | PR-03             |
| 3.12 | Commit: resumable, deduplicated by normalized geometry hash                                                                                   | Re-running commit creates no duplicates                       | PR-03             |

## Phase 3–5 (S7–S12)

| #    | Task                                                                                                       | Done when                                    | Spec           |
|------|------------------------------------------------------------------------------------------------------------|----------------------------------------------|----------------|
| 3.13 | <code>soil_tests</code> with method, depth, date, units, source                                            | W17 data path works (feeds onboarding)       | PR-13 data     |
| 3.14 | Consent records per purpose                                                                                | Consent withdrawal blocks monitoring         | §05 commercial |
| 3.15 | New onboarding API replacing the 9 legacy steps                                                            | Frontend onboarding runs on <code>/v1</code> | —              |
| 3.16 | <code>tools/migrate-legacy</code> : farmers → users/memberships, farms → farms/fields, crops → crop cycles | Dry-run report + idempotent run              | —              |
| 3.17 | S10: rehearse migration twice on a copy of the live DB                                                     | Row counts and spot checks signed off        |                |
| 3.18 | S11–S12: migrate into staging, then production; support pilot onboarding                                   | Pilot farms visible with history             |                |

## P4 — Backend: operational workflows

**Mission:** turn observations into action: weather, alerts, scouting, activities, reports, notifications — plus the commercial and privacy controls.

### Phase 0–1 (S1–S3)

| #   | Task                                                                                                                          | Done when                              | Spec         |
|-----|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|--------------|
| 4.1 | <code>weather.Provider</code> interface; Open-Meteo implementation (provider, issue time, valid interval, units, attribution) | Adapter fixtures incl. 429/5xx/timeout | §06.5, PR-07 |
| 4.2 | Alert rule specification document with P2 (thresholds from §04.5, suppression reasons)                                        | Approved                               | PR-08        |
| 4.3 | <code>weather_samples</code> storage + field-grid binding (no per-field copies of identical data)                             | T14 passes (timezone/DST)              | §07.4, W11   |

| #   | Task                                        | Done when                        | Spec  |
|-----|---------------------------------------------|----------------------------------|-------|
| 4.4 | entitlements , usage_buckets , usage_ledger | Plan limits stored and versioned | PR-25 |

## Phase 2 (S4–S6)

| #   | Task                                                                                          | Done when                                             | Spec       |
|-----|-----------------------------------------------------------------------------------------------|-------------------------------------------------------|------------|
| 4.5 | Atomic admission check before costly work (analysis, report)                                  | Over-limit → rejected before job creation             | PR-25, W28 |
| 4.6 | alerts : episodes, dedupe, severity, evidence date, factual reason                            | Duplicate events do not create a second episode (T06) | W12        |
| 4.7 | Data-gap alerts: "no new image" ≠ "processing failed"                                         | T30 passes                                            | PR-08      |
| 4.8 | Alert triage: acknowledge, dismiss with reason code ( false_positive ), convert to scout task | W13 demoed                                            | W13        |

## Phase 3 (S7–S9)

| #    | Task                                                                                              | Done when     | Spec              |
|------|---------------------------------------------------------------------------------------------------|---------------|-------------------|
| 4.9  | scout_tasks , inspections with state machine                                                      | W14 demoed    | \$03 state tables |
| 4.10 | Offline sync endpoint: offline_operations , operation ID dedupe, base_version conflicts           | T15, T16 pass | W15               |
| 4.11 | Photo evidence via quarantine → validated → ready                                                 | T17 passes    | \$06.4            |
| 4.12 | activities : planned vs actual dates, costs in integer minor units, no mixed currency totals      | T18 passes    | W16, PR-10        |
| 4.13 | reports worker: CSV and PDF, snapshot cutoff, reauthorize on download                             | T20 passes    | W21, PR-11        |
| 4.14 | notification_preferences , notifications , delivery_attempts ; quiet hours; unknown-outcome state | W23 demoed    | PR-12             |
| 4.15 | privacy_requests : export and erasure using P1's job framework                                    | W24 demoed    | PR-26             |

## Phase 4–5 (S10–S12)

| #    | Task                                                        | Done when                    |
|------|-------------------------------------------------------------|------------------------------|
| 4.16 | T25 (Redis loss keeps quota/idempotency) and T30 regression | Pass                         |
| 4.17 | Google Weather API adapter behind the same interface (S11)  | Staging uses it              |
| 4.18 | Production notification (SMS/email) provider wiring (S12)   | Pilot receives notifications |

## P5 — Frontend lead

**Mission:** a static Next.js app that never shows missing data as healthy and never lets UI hide authorization problems.

### Phase 0–1 (S1–S3)

| #   | Task                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Done when                                 | Spec   |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------|
| 5.1 | <code>apps/web</code> : Next.js, <code>output: 'export'</code> , <code>trailingSlash</code> , unoptimized images                                                                                                                                                                                                                                                                                                                                                                                                                   | Build emits only static files             | \$08.1 |
| 5.2 | Routes: <code>/login/</code> , <code>/app/</code> , <code>/app/fields/</code> , <code>/app/field/</code> , <code>/app/seasons/</code> , <code>/app/scouting/</code> , <code>/app/task/</code> , <code>/app/activities/</code> , <code>/app/weather/</code> , <code>/app/alerts/</code> , <code>/app/zoning/</code> , <code>/app/advice/</code> , <code>/app/reports/</code> , <code>/app/integrations/</code> , <code>/app/team/</code> , <code>/app/settings/</code> , <code>/oauth/callback/</code> (P1+ routes as placeholders) | Deep links work from static hosting (T28) | \$08.2 |
| 5.3 | Design tokens, components, map library decision (MapLibre) as ADR                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ADR merged                                | \$08.3 |
| 5.4 | Typed API client generated from <code>contracts/openapi.json</code> ; TanStack Query with keys incl. tenant + scope revision                                                                                                                                                                                                                                                                                                                                                                                                       | Tenant switch clears caches               | \$08.3 |
| 5.5 | Sign-in with Firebase emulator, token refresh via SDK, tenant selector                                                                                                                                                                                                                                                                                                                                                                                                                                                             | W01 demoed                                | \$08.4 |
| 5.6 | Field list + map, draw/edit polygon, validation errors displayed from <code>problem+json</code>                                                                                                                                                                                                                                                                                                                                                                                                                                    | W04, W06 demoed                           | PR-02  |

### Phase 2 (S4–S6)

| #    | Task                                                                                                              | Done when                          | Spec           |
|------|-------------------------------------------------------------------------------------------------------------------|------------------------------------|----------------|
| 5.7  | Scene timeline and index chart: forecast vs observation, missing $\neq$ zero                                      | Visual check against fixtures      | \$01.4         |
| 5.8  | Field detail: crop cycle, history, quality class, source, sensing time, resolution, processing version            | PR-05 acceptance visible on screen | PR-05          |
| 5.9  | Map tiles through the gateway with bearer header; legend with textual numbers, never colour alone                 | Accessibility check passes         | \$02.6, \$08.6 |
| 5.10 | Compare dates, natural-colour check, contrast mode shows actual min/max                                           | W10 demoed                         | \$04.4         |
| 5.11 | Home work list: urgent tasks, new evidence, data gaps; alerts screen; weather screen                              | PR-07/08 visible                   | \$01.4         |
| 5.12 | Port useful pieces of legacy <code>HeatmapLayer.jsx</code> , <code>Legend.jsx</code> , <code>colorUtils.js</code> | —                                  | —              |

### Phase 3–4 (S7–S10)

| #    | Task                                              | Done when  | Spec  |
|------|---------------------------------------------------|------------|-------|
| 5.13 | Team and invitations UI, protected owner controls | W03 demoed | PR-01 |

| #    | Task                                                                            | Done when                              | Spec   |
|------|---------------------------------------------------------------------------------|----------------------------------------|--------|
| 5.14 | Onboarding UI on the new API (reuse the step UX from <code>pages/steps</code> ) | New farmer onboards end to end         | —      |
| 5.15 | Accessibility WCAG 2.2 AA on primary flows; RTL-safe layout                     | Automated + manual checks pass         | \$08.6 |
| 5.16 | English/Hindi via message keys                                                  | All strings externalized               | \$01.4 |
| 5.17 | Performance: lazy-load map/charts, virtualize lists, abort superseded requests  | Shell usable < 3 s on mid-range device | \$01.6 |
| 5.18 | Delete <code>frontend/</code> (S10)                                             | Repo builds without it                 | —      |
| 5.19 | S11: Firebase Hosting config, cache headers, CSP (with P6)                      | Headers verified in staging            | \$08.5 |

## P6 — Frontend + QA/DevEx, cloud lead in Phase 5

**Mission:** prove every claim with tests, build the offline workflow UI, then take the product to the cloud.

### Phase 0–1 (S1–S3)

| #   | Task                                                                                                                                                                    | Done when                 | Spec     |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------|
| 6.1 | GitHub Actions CI: <code>go vet</code> , <code>staticcheck</code> , <code>go test -race</code> , migrations up on PostGIS container, web lint/build, OpenAPI validation | Required on every PR      | \$12.1   |
| 6.2 | Keep legacy <code>make contract</code> running in CI until S10                                                                                                          | Legacy stays green        | —        |
| 6.3 | Test harness: Go integration tests with ephemeral PostGIS, Playwright, k6                                                                                               | Sample test in each layer | \$12.1   |
| 6.4 | Tenant isolation suite (with P1): every endpoint × wrong tenant / revoked member                                                                                        | T01, T02 in CI            | T01, T02 |
| 6.5 | Tests for untested packages still used after migration ( <code>jwtutil</code> , <code>firebase</code> )                                                                 | Coverage no longer 0%     | AUDIT M9 |
| 6.6 | T03 (100 repeated keys) and T08 geometry tests in CI                                                                                                                    | Pass                      | T03, T08 |

### Phase 2 (S4–S6)

| #    | Task                                                                     | Done when                                      |
|------|--------------------------------------------------------------------------|------------------------------------------------|
| 6.7  | E2E against the <b>exported static build</b> , not <code>next dev</code> | Sign-in → draw field → first observation → map |
| 6.8  | T09, T10, T11, T12, T13, T14 as automated tests with P2/P4               | Pass in CI                                     |
| 6.9  | Benchmark harness for P2's 1,000-field run                               | Repeatable report                              |
| 6.10 | Import wizard UI: upload, column mapping, dry-run table, commit progress | W05 demoed                                     |

### Phase 3–4 (S7–S10)

| #    | Task                                                                                                              | Done when                      | Spec   |
|------|-------------------------------------------------------------------------------------------------------------------|--------------------------------|--------|
| 6.11 | Offline store in IndexedDB: assigned tasks, selected outlines, drafts; 7-day retention; purge control             | Works in airplane mode         | \$08.4 |
| 6.12 | Sync queue: operation_id, device_id, base_version; foreground retry without Background Sync                       | T15 passes on device           | \$08.4 |
| 6.13 | Conflict screen: server version vs local draft, nothing lost                                                      | T15, T16 pass                  | W15    |
| 6.14 | Scouting UI (assignment, map point, photos, submit), activities UI, reports UI, settings/notifications/privacy UI | W14, W16, W21, W23, W24 demoed | —      |
| 6.15 | Security regression: T17, T20, SSRF, malicious files                                                              | Pass                           | \$09.7 |
| 6.16 | Local load test (baseline + burst)                                                                                | Report with p50/p95/p99        | \$12.4 |
| 6.17 | Release checklist and full regression (S10)                                                                       | Signed                         | \$11.6 |

### Phase 5 (S11–S12) — cloud lead

| #    | Task                                                                                                                                                                                                                                                                 | Done when                                       | Spec         |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|--------------|
| 6.18 | Terraform modules: projects, VPC, Cloud NAT, Cloud SQL (HA, PITR, PostGIS), Memorystore, GCS buckets, Pub/Sub, Cloud Tasks, Secret Manager, Cloud Run (api/worker/tile/maintenance, one service account each), LB + Cloud Armor, Firebase Hosting, Identity Platform | terraform plan clean for staging and production | \$02, \$10.6 |
| 6.19 | Workload Identity Federation for CI; <b>no JSON key files</b> in any environment                                                                                                                                                                                     | serviceAccountKey.json unused in staging/prod   | \$02.3       |
| 6.20 | CI/CD: build image → migration Cloud Run job → deploy staging → smoke → manual prod promotion with traffic split → rollback by revision                                                                                                                              | Demonstrated rollback                           | \$10.6       |
| 6.21 | Monitoring: SLO dashboards, alerts, structured logs, traces; budget alerts for EE and AI                                                                                                                                                                             | Alerts fire in a test                           | \$10.3–5     |
| 6.22 | Restore drill and cross-region DR test                                                                                                                                                                                                                               | Measured RPO/RTO recorded; T27 passes           | \$10.7       |
| 6.23 | T26 (connection budget), T28, T29 in staging                                                                                                                                                                                                                         | Pass                                            | \$12.2       |
| 6.24 | Pilot rollout (20 farms) with runbooks                                                                                                                                                                                                                               | Go/no-go signed                                 | \$11.6       |

## 8. Handoffs between people

| From → To       | What                                                    | Due     |
|-----------------|---------------------------------------------------------|---------|
| P1 → all        | Local stack, AGENTS.md , OpenAPI draft + mock           | End S1  |
| P1 → all        | Frozen OpenAPI, tx helper, auth verifier, RLS           | End S2  |
| P3 → P1         | Legacy → new table mapping                              | End S1  |
| P1 → P2, P4     | Jobs + leases, outbox                                   | End S3  |
| P3 → P2         | Fields + boundary versions (FK target for observations) | End S3  |
| P1 → P2, P3, P4 | queue.Dispatcher , storage.Bucket                       | End S4  |
| P2 → P5         | Observation + latest-projection endpoints               | End S5  |
| P2 → P5         | Tile gateway                                            | End S6  |
| P2 → P4         | Change-detection events                                 | End S6  |
| P4 → P5         | Alerts and weather endpoints                            | End S6  |
| P3 → P6         | Import dry-run/commit endpoints                         | S5 / S6 |
| P4 → P6         | Scouting, sync, activities endpoints                    | End S8  |
| P4 → P6         | Reports, notifications, privacy endpoints               | End S9  |
| P1, P3 → P6     | Adapter swap list + migration plan for staging          | End S10 |

A handoff is complete only when the endpoint is in the frozen contract, implemented, tested, and demoed on the local stack.

## 9. Acceptance scenarios by owner

Scenarios from spec §12.2 that apply to P0. The owner writes the implementation; P6 makes sure the test runs in CI.

| Scenario                                                    | Owner   | Sprint |
|-------------------------------------------------------------|---------|--------|
| T01 cross-tenant access denied everywhere                   | P1      | S2     |
| T02 revoked membership denied on next request               | P1      | S2     |
| T03 same key ×100 → one effect; changed body conflicts      | P1      | S3     |
| T04 crash after commit replays response                     | P1      | S3     |
| T05 crash after export before commit reconciles once        | P2      | S5     |
| T06 duplicate/out-of-order events do not regress version    | P1 / P4 | S6     |
| T07 failed dispatch retried without losing work             | P1      | S4     |
| T08 invalid geometries return precise errors                | P3      | S3     |
| T09 boundary edit during analysis keeps provenance          | P2 / P3 | S5     |
| T10 all-cloud scene → insufficient, never NDVI 0            | P2      | S5     |
| T11 20 m resolution stays explicit                          | P2      | S6     |
| T12 scaled reflectance formulas, masked denominators        | P2      | S3     |
| T13 change detection on common valid area                   | P2      | S6     |
| T14 weather times survive timezone/DST                      | P4      | S6     |
| T15 offline duplicate → one inspection; draft kept          | P4 / P6 | S8     |
| T16 revoked scout cannot sync to old field                  | P4      | S8     |
| T17 photo upload cannot bypass scan                         | P4      | S8     |
| T18 activity completion needs actual dates, units, currency | P4      | S8     |
| T20 report download blocked after revocation                | P4      | S9     |
| T25 Redis loss keeps quota/idempotency                      | P1 / P4 | S10    |
| T26 DB connection budget during deploy and burst            | P6      | S12    |

| Scenario                                       | Owner   | Sprint |
|------------------------------------------------|---------|--------|
| T27 restore applies erasure ledger first       | P6 / P1 | S12    |
| T28 static deep links work without Next server | P5      | S2     |
| T29 logs/metrics carry no private fields       | P6      | S12    |
| T30 provider outage shows stale/degraded state | P4 / P5 | S6     |

T19, T21–T24 belong to post-pilot features (§13).

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## 10. Definition of done

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A task is done only when **all** apply (spec §11.6):

- ☐ Code reviewed by the folder owner (and P1 for anything touching tenancy, auth or transactions)
- ☐ Unit tests including negative cases; integration test against real PostGIS where SQL is involved
- ☐ OpenAPI updated (if contract-affecting) and approved by P1
- ☐ Migration tested up from empty DB
- ☐ Structured logs with request ID; no private fields logged
- ☐ Permission and retention implications written in the PR description
- ☐ UI work: keyboard accessible, text alternative for colour, loading ≠ empty ≠ error states
- ☐ Demoed on the local stack at the Friday review

A demo alone is not done.

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## 11. Working agreements

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### Branching and PRs

- Trunk-based: short branches off `main`, merged daily.
- PRs under ~400 changed lines where possible.
- CODEOWNERS enforces §4 ownership; CI is required to merge.

### Rituals

| When          | What                                     | Who                 |
|---------------|------------------------------------------|---------------------|
| Daily, 15 min | Stand-up: yesterday, today, blocked      | All                 |
| Tue & Thu     | Dependency check against §8 handoffs     | P1 + anyone blocked |
| Friday        | Demo on the local stack; scope decisions | All                 |
| Sprint end    | Review + retro; update this document     | All                 |

### `AGENTS.md` rules (Antigravity)

1. Only edit folders owned by the person running the agent (see §4).
2. Never edit `contracts/`, `migrations/` or `internal/platform/` without P1 review.
3. Every repository query uses the tenant transaction helper; no unscoped `GetByID`.
4. Every new Earth Engine call ships with a recorded fixture.
5. Masked or missing values serialize as `absent/ null`, never `0`.
6. No new third-party dependency without an ADR.
7. Run the full test target before opening a PR.
8. Never commit `.env`, key files or real farmer data.

### Metrics tracked each sprint

Cycle time, blocked days, escaped defects, open handoffs, scope changes.

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## 12. Risks

| Risk                                                                                     | Owner   | Trigger                            | Mitigation                                                                                  |
|------------------------------------------------------------------------------------------|---------|------------------------------------|---------------------------------------------------------------------------------------------|
| Contract slips past S2                                                                   | P1      | OpenAPI not frozen end S2          | P1 has no other work in S1–S2; frontend uses mock                                           |
| Cloud-last surprises (Cloud Run CPU suspension, private networking, EE commercial quota) | P6 / P2 | Staging differs from local         | All adapters behind interfaces; Phase 5 budgeted at 4 weeks; EE quota verified first in S11 |
| EE batch export complexity from Go                                                       | P2      | Export/reconcile not working by S5 | Start with <code>SubmitStatistics</code> only; exports for large fields follow              |
| Small fields / cloud cover make data insufficient                                        | P2      | Many pilot fields "insufficient"   | Quality classes shown honestly; scouting fallback                                           |
| Legacy data loss                                                                         | P3      | Migration mismatch                 | Two rehearsals on a copy; idempotent script; row-count checks                               |
| Offline conflicts lose drafts                                                            | P6 / P4 | T15 failures                       | Operation ledger, explicit conflict UI, device tests                                        |
| Agents overwriting each other                                                            | P1      | Merge conflicts across folders     | CODEOWNERS, <code>AGENTS.md</code> , small daily PRs                                        |
| Scope creep into AI/VRA                                                                  | All     | Requests before pilot              | \$13 list is closed until the pilot starts                                                  |
| One person out for a sprint                                                              | All     | Absence                            | Pair on critical-path items (P1↔P3, P2↔P4, P5↔P6)                                           |

### 13. After the pilot

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Not in this plan. Candidate owners once the pilot runs:

| Requirement   | Feature                                                                                          | Likely owner |
|---------------|--------------------------------------------------------------------------------------------------|--------------|
| PR-13 – PR-15 | Soil interpretation, growth stages, water stress                                                 | P2 + P4      |
| PR-16         | VRA zoning and prescriptions (W18, W19, T19)                                                     | P2           |
| PR-17         | AI assistant with citations and review (W20, T22, T23) — builds on <code>internal/chatbot</code> | P4           |
| PR-18         | John Deere, sensors, OAuth (W22, T21)                                                            | P3           |
| PR-19 – PR-22 | Yield, disease, classification, damage, commercial imagery (T24)                                 | P2 + ML hire |
| PR-23 – PR-24 | Enterprise branding, imagery exploration                                                         | P5           |